

ECON3200 Tutorial 9 Recap and Solutions

Open Economies, Monetary Policy Rules and a Trade-off

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1. Recap Lectures 6, 7, 8 and 9

1.1. Lecture 6: Money in the Short Run

Lecture 6 explains why money is non-neutral in the short run. The key reason is price and wage rigidities. Because prices and wages do not adjust immediately, monetary policy can affect real variables such as output and employment.

A contractionary monetary policy shock usually leads to

$$\hat{i}_t \uparrow \Rightarrow \hat{y}_t \downarrow \Rightarrow \hat{\pi}_t \downarrow \text{ slowly } \Rightarrow \hat{m}_t \downarrow.$$

The negative relationship between money supply and nominal interest rates is called the liquidity effect. Variables are written as percentage deviations around the steady state. For example, $\hat{u}_t < 0$ means u_t is below steady state, while $\hat{u}_t > 0$ means u_t is above steady state.

1.2. Lecture 7: Basic New Keynesian Model and Monetary Policy Shock

The basic New Keynesian model is summarized by the NK IS curve, the NK Phillips curve, and the monetary policy rule:

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t, \quad (1.NK^*)$$

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t, \quad (2.NK^*)$$

$$\hat{i}_t = \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t. \quad (3.NK^*)$$

The output gap is

$$\hat{x}_t = \hat{y}_t - \hat{y}_t^f.$$

Here $\hat{y}_t$ is actual output and $\hat{y}_t^f$ is flexible-price output. A monetary policy shock enters through

$$v_t = \rho_v v_{t-1} + \varepsilon_t.$$

A contractionary monetary policy shock means $\varepsilon_t > 0$. Then

$$\varepsilon_t \uparrow \Rightarrow v_t \uparrow \Rightarrow \hat{i}_t \uparrow.$$

The real interest rate is

$$\hat{r}_t = \hat{i}_t - E_t[\hat{\pi}_{t+1}].$$

Since prices are sticky, expected inflation does not rise one-for-one with the nominal interest rate. Hence,

$$\hat{i}_t \uparrow \Rightarrow \hat{r}_t \uparrow \Rightarrow \hat{x}_t \downarrow \Rightarrow \hat{\pi}_t \downarrow.$$

Using the output-gap definition, a monetary policy shock mainly lowers actual output $\hat{y}_t$, while flexible-price output $\hat{y}_t^f$ does not move much because fundamentals such as technology do not change. Therefore,

$$\hat{y}_t \downarrow, \quad \hat{y}_t^f \text{ unchanged} \Rightarrow \hat{x}_t = \hat{y}_t - \hat{y}_t^f \downarrow.$$

1.3. Lecture 8: Technology Shock, Spending Shock and Cost-Push Shock

1. Technology shock

Productivity follows

$$a_t = \rho_a a_{t-1} + \omega_t, \quad \hat{z}_t = a_t.$$

Flexible-price output is

$$\hat{y}_t^f = \left(\frac{1 + \phi}{\sigma + \phi} \right) \hat{z}_t.$$

A positive technology shock means $\omega_t > 0$, so

$$\omega_t > 0 \Rightarrow \hat{z}_t \uparrow \Rightarrow \hat{y}_t^f \uparrow.$$

The key point is that a positive technology shock increases both actual output and flexible-price output, but flexible-price output increases more. Therefore,

$$\hat{x}_t = \hat{y}_t - \hat{y}_t^f \downarrow.$$

Then the Phillips curve implies

$$\hat{x}_t \downarrow \Rightarrow \hat{\pi}_t \downarrow.$$

Employment is

$$\hat{l}_t = \hat{y}_t - \hat{z}_t.$$

If productivity rises more than actual output, employment falls:

$$\hat{z}_t \uparrow \text{ more than } \hat{y}_t \Rightarrow \hat{l}_t \downarrow.$$

2. Spending / demand shock

Lecture 8 introduces a spending shock:

$$g_t = \rho_g g_{t-1} + \omega_t^g.$$

Flexible-price output becomes

$$\hat{y}_t^f = \left(\frac{1 + \phi}{\sigma + \phi} \right) \hat{z}_t + \left(\frac{1}{\sigma + \phi} \right) g_t.$$

This is because g_t enters household preferences and affects desired consumption and labour supply. The IS curve becomes

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t + u_t^g, \quad (1.NK^{**})$$

where

$$u_t = E_t[\hat{y}_{t+1}^f] - \hat{y}_t^f, \quad u_t^g = \frac{1}{\sigma} (g_t - E_t[g_{t+1}]).$$

A positive spending shock increases current demand:

$$g_t \uparrow \Rightarrow u_t^g \uparrow \Rightarrow \hat{x}_t \uparrow \Rightarrow \hat{\pi}_t \uparrow \Rightarrow \hat{i}_t \uparrow.$$

3. Cost-push shock

With a cost-push shock, the model becomes

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t, \quad (1.NK^{***})$$

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t + \varepsilon_t^w, \quad (2.NK^{***})$$

$$\hat{i}_t = \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t. \quad (3.NK^{***})$$

The cost-push shock follows

$$\varepsilon_t^w = \rho_w \varepsilon_{t-1}^w + \varepsilon_t^o.$$

A positive cost-push shock directly raises inflation:

$$\varepsilon_t^w \uparrow \Rightarrow \hat{\pi}_t \uparrow.$$

Then the Central Bank raises the nominal interest rate:

$$\hat{\pi}_t \uparrow \Rightarrow \hat{i}_t \uparrow \Rightarrow \hat{x}_t \downarrow.$$

Therefore, cost-push shocks create a trade-off:

$$\varepsilon_t^w \uparrow \Rightarrow \hat{\pi}_t \uparrow \quad \text{and/or} \quad \hat{x}_t \downarrow.$$

2. Lecture 9 Recap

2.1. Open Economy

In a small open economy, CPI inflation and the output gap are affected by the terms of trade:

$$\hat{s}_t = \hat{p}_t^f - \hat{p}_t^h, \quad \Delta \hat{s}_t = \hat{s}_t - \hat{s}_{t-1}.$$

Here, $\hat{s}_t$ is the terms of trade. It measures the relative price of foreign goods compared with home goods. The notation $\hat{p}_t^f$ means the price of foreign goods, and $\hat{p}_t^h$ means the price of home goods. Therefore, if $\hat{s}_t$ rises, foreign goods become relatively more expensive compared with home goods. If $\hat{s}_t$ falls, foreign goods become relatively cheaper.

The term $\Delta \hat{s}_t$ means the change in the terms of trade:

$$\Delta \hat{s}_t = \hat{s}_t - \hat{s}_{t-1}.$$

So $\Delta \hat{s}_t > 0$ means foreign goods become relatively more expensive this period, while $\Delta \hat{s}_t < 0$ means foreign goods become relatively cheaper this period.

The parameter $\gamma \in (0, 1)$ measures the weight of foreign-produced goods in domestic consumption. If $\gamma = 0$, the economy is closed because households consume only home goods. A larger γ means the economy is more open because foreign goods have a larger weight in the consumption basket.

CPI inflation is

$$\hat{\pi}_t^c = \hat{\pi}_t^h + \gamma \Delta \hat{s}_t.$$

In this equation, $\hat{\pi}_t^c$ is CPI inflation, meaning the inflation rate of the whole consumption basket. The superscript c stands for consumption. The term $\hat{\pi}_t^h$ is home-goods inflation, meaning inflation for domestically produced goods. The term $\gamma \Delta \hat{s}_t$ captures the effect of changes in foreign goods prices relative to home goods prices.

Therefore, CPI inflation depends on two parts:

$$\text{CPI inflation} = \text{home-goods inflation} + \text{terms-of-trade effect}.$$

If foreign goods become relatively cheaper, then

$$\Delta \hat{s}_t < 0.$$

Because $\gamma > 0$, this makes

$$\gamma \Delta \hat{s}_t < 0,$$

so CPI inflation falls:

$$\hat{\pi}_t^c \downarrow.$$

If foreign goods become relatively more expensive, then

$$\Delta \hat{s}_t > 0,$$

so CPI inflation rises:

$$\hat{\pi}_t^c \uparrow.$$

The open-economy system is

$$\begin{aligned}\hat{s}_t &= (\hat{i}_t^* - E_t[\hat{\pi}_{t+1}^{c*}]) - (\hat{i}_t - E_t[\hat{\pi}_{t+1}^c]) + E_t[\hat{s}_{t+1}], \\ \hat{x}_t &= E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}^c] \right) - \frac{\gamma\psi}{\sigma} E_t[\Delta \hat{s}_{t+1}] + u_t, \\ \hat{\pi}_t^c &= \beta E_t[\hat{\pi}_{t+1}^c] + \tilde{\kappa}_o \hat{x}_t - \beta \gamma E_t[\Delta \hat{s}_{t+1}] + \gamma \Delta \hat{s}_t, \\ \hat{i}_t &= \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t.\end{aligned}$$

The first equation shows that the terms of trade depend on real interest rate differentials. The domestic real interest rate is

$$\hat{i}_t - E_t[\hat{\pi}_{t+1}^c],$$

while the foreign real interest rate is

$$\hat{i}_t^* - E_t[\hat{\pi}_{t+1}^{c*}].$$

If the domestic real interest rate rises relative to the foreign real interest rate, domestic assets become more attractive. This leads to an appreciation of the domestic currency. In the lecture notation, this means the terms of trade fall:

$$\hat{s}_t \downarrow.$$

The second equation is the open-economy IS curve. It shows that the output gap depends on the domestic real interest rate and expected changes in the terms of trade. A higher real interest rate reduces current demand:

$$\hat{i}_t - E_t[\hat{\pi}_{t+1}^c] \uparrow \Rightarrow \hat{x}_t \downarrow.$$

The extra open-economy term,

$$-\frac{\gamma\psi}{\sigma} E_t[\Delta \hat{s}_{t+1}],$$

shows that expected movements in the terms of trade also affect demand.

The third equation is the open-economy Phillips curve. It says CPI inflation depends on expected future CPI inflation, the output gap, and current and expected changes in the terms of trade. A lower output gap reduces inflation:

$$\hat{x}_t \downarrow \Rightarrow \hat{\pi}_t^c \downarrow.$$

A fall in the terms of trade also reduces CPI inflation:

$$\Delta \hat{s}_t < 0 \Rightarrow \hat{\pi}_t^c \downarrow.$$

A contractionary monetary policy shock in an open economy works through both the standard demand channel and the exchange-rate / terms-of-trade channel:

$$\hat{i}_t \uparrow \Rightarrow \hat{r}_t \uparrow \Rightarrow \hat{x}_t \downarrow \Rightarrow \hat{\pi}_t^c \downarrow.$$

This is the standard demand channel. A higher interest rate raises the real interest rate, reduces consumption and demand, lowers the output gap, and reduces inflation.

In an open economy, there is also an extra channel:

$$\hat{i}_t \uparrow \Rightarrow \hat{s}_t \downarrow \Rightarrow \Delta \hat{s}_t < 0 \Rightarrow \hat{\pi}_t^c \downarrow.$$

The intuition is that a higher domestic interest rate makes domestic assets more attractive, causing the domestic currency to appreciate. This makes foreign goods relatively cheaper, so the terms of trade fall. Since CPI inflation includes foreign goods, the fall in the terms of trade lowers CPI inflation directly.

Putting both channels together:

$$\hat{i}_t \uparrow \Rightarrow \hat{r}_t \uparrow \Rightarrow \hat{x}_t \downarrow \Rightarrow \hat{\pi}_t^c \downarrow,$$

and

$$\hat{i}_t \uparrow \Rightarrow \hat{s}_t \downarrow \Rightarrow \hat{\pi}_t^c \downarrow.$$

Therefore, monetary policy has a stronger effect in an open economy than in a closed economy. The effect is stronger when γ is larger because foreign goods have a larger weight in CPI inflation:

$$\gamma \uparrow \Rightarrow \text{terms-of-trade channel becomes stronger.}$$

2.2. Monetary Policy Rules and Stability

Lecture 9 compares three policy rules:

$$\hat{i}_t = \rho_i \hat{i}_{t-1} + v_t. \quad (3A.NK^*)$$

This rule does not respond to inflation or the output gap. If expected inflation rises, $\hat{i}_t$ does not respond, so

$$\hat{r}_t = \hat{i}_t - E_t[\hat{\pi}_{t+1}] \downarrow \Rightarrow \hat{x}_t \uparrow \Rightarrow \hat{\pi}_t \uparrow.$$

So expectations can become self-fulfilling.

$$\hat{i}_t = \delta \hat{\pi}_t + v_t. \quad (3B.NK^*)$$

This rule responds to inflation. The economy has a unique equilibrium if

$$\delta > 1.$$

This is the Taylor Principle.

$$\hat{i}_t = \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t. \quad (3C.NK^*)$$

This rule responds to both inflation and the output gap, so it helps stabilize the economy.

2.3. Trade-off in Monetary Policy Design

With a cost-push shock, the Phillips curve can be written as

$$\hat{\pi}_t = \tilde{\kappa} \sum_{k=0}^{\infty} \beta^k E_t[\hat{x}_{t+k}] + \varepsilon_t^w. \quad (2TO.NK^*)$$

A positive cost-push shock directly raises inflation:

$$\varepsilon_t^w > 0 \Rightarrow \hat{\pi}_t \uparrow.$$

To reduce inflation, the Central Bank must create a negative current or expected future output gap:

$$\hat{x}_{t+k} < 0.$$

Thus,

stabilize inflation $\Rightarrow$ lower output gap, stabilize output $\Rightarrow$ higher inflation.

3. Tutorial 9 Full Questions and Solutions

3.1. Exercise 9.1

Question

Consider

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t, \quad (1.NK^*)$$

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t, \quad (2.NK^*)$$

$$\hat{i}_t = \rho_i \hat{i}_{t-1} + v_t, \quad (3A.NK^*)$$

with

$$\tilde{\kappa} = (\sigma + \phi) \frac{(1 - \kappa)(1 - \beta\kappa)}{\kappa}.$$

Assume the economy is at steady state in period -1 , $u_t = v_t = 0$, $E_0[\hat{\pi}_1] = 0.01$, $E_0[\hat{x}_1] = 0$, $\beta = 0.9$, $\sigma = 1$, and $\kappa = 0.5$. Calculate the effect on $\hat{x}_0$ and $\hat{\pi}_0$.

Given:

$$E_0[\hat{\pi}_1] = 0.01, \quad E_0[\hat{x}_1] = 0, \quad u_0 = v_0 = 0, \quad \beta = 0.9, \quad \sigma = 1, \quad \tilde{\kappa} = 1.1.$$

Step 1: Find $\hat{i}_0$.

Rule (3A.NK*):

$$\hat{i}_t = \rho_i \hat{i}_{t-1} + v_t.$$

Since the economy starts at steady state and $v_0 = 0$,

$$\hat{i}_0 = \underline{\hspace{2cm}}.$$

Step 2: Use the IS curve.

$$\hat{x}_0 = E_0[\hat{x}_1] - \frac{1}{\sigma} \left(\hat{i}_0 - E_0[\hat{\pi}_1] \right) + u_0.$$

Substitute:

$$\hat{x}_0 = 0 - \frac{1}{1} (\underline{\hspace{2cm}} - 0.01) + 0 = \underline{\hspace{2cm}}.$$

Step 3: Use the Phillips curve.

$$\hat{\pi}_0 = \beta E_0[\hat{\pi}_1] + \tilde{\kappa} \hat{x}_0.$$

Substitute:

$$\hat{\pi}_0 = 0.9(0.01) + 1.1(\underline{\hspace{2cm}}) = \underline{\hspace{2cm}}.$$

Final answers:

$$\hat{x}_0 = \underline{\hspace{2cm}}, \quad \hat{\pi}_0 = \underline{\hspace{2cm}}.$$

Intuition: Expected inflation rises, but the nominal interest rate does not respond. Hence the real interest rate falls, demand rises, the output gap rises, and inflation rises.

Solution

For these values,

$$\tilde{\kappa} = 1.1.$$

Since the economy is at steady state and there are no shocks,

$$\hat{i}_0 = 0.$$

Using the IS curve:

$$\hat{x}_0 = 0 - \frac{1}{1}(0 - 0.01) = 0.01.$$

Using the Phillips curve:

$$\hat{\pi}_0 = 0.9(0.01) + 1.1(0.01) = 0.009 + 0.011 = 0.02.$$

Therefore,

$$\boxed{\hat{x}_0 = 0.01, \quad \hat{\pi}_0 = 0.02.}$$

Explanation

Expected inflation rises, but the nominal interest rate does not respond. Hence,

$$\hat{r}_0 = \hat{i}_0 - E_0[\hat{\pi}_1] = 0 - 0.01 = -0.01.$$

A lower real interest rate raises demand, so $\hat{x}_0 \uparrow$. Then the Phillips curve gives $\hat{\pi}_0 \uparrow$. This is self-fulfilling expectations.

3.2. Exercise 9.2

Question

Consider

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t, \quad (1.NK^*)$$

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t, \quad (2.NK^*)$$

$$\hat{i}_t = \delta \hat{\pi}_t + v_t, \quad (3B.NK^*)$$

with

$$\tilde{\kappa} = (\sigma + \phi) \frac{(1 - \kappa)(1 - \beta\kappa)}{\kappa}.$$

Assume the economy is at **steady state in period -1** , $u_t = v_t = 0$, $E_0[\hat{\pi}_1] = 0.01$, $E_0[\hat{x}_1] = 0$, $\beta = 0.9$, $\sigma = 1$, $\kappa = 0.5$, and $\delta = 1$. Calculate the effect on $\hat{x}_0$ and $\hat{\pi}_0$.

Given:

$$E_0[\hat{\pi}_1] = 0.01, \quad E_0[\hat{x}_1] = 0, \quad u_0 = v_0 = 0, \quad \beta = 0.9, \quad \sigma = 1, \quad \tilde{\kappa} = 1.1, \quad \delta = 1.$$

Step 1: Use the monetary policy rule.

Rule (3B.NK*):

$$\hat{i}_t = \delta \hat{\pi}_t + v_t.$$

Since $\delta = 1$ and $v_0 = 0$,

$$\hat{i}_0 = \hat{\pi}_0.$$

Step 2: Use the IS curve.

$$\hat{x}_0 = - \left(\hat{i}_0 - E_0[\hat{\pi}_1] \right).$$

Substitute $\hat{i}_0 = \hat{\pi}_0$:

$$\hat{x}_0 = - (\hat{\pi}_0 - 0.01) = 0.01 - \hat{\pi}_0.$$

Step 3: Use the Phillips curve.

$$\hat{\pi}_0 = 0.9(0.01) + 1.1\hat{x}_0.$$

Substitute $\hat{x}_0 = 0.01 - \hat{\pi}_0$:

$$\hat{\pi}_0 = 0.9(0.01) + 1.1(0.01 - \hat{\pi}_0).$$

Simplify:

$$\hat{\pi}_0 = 0.009 + 0.011 - 1.1\hat{\pi}_0.$$

$$\hat{\pi}_0 = 0.02 - 1.1\hat{\pi}_0.$$

Move terms:

$$2.1\hat{\pi}_0 = 0.02.$$

Solve:

$$\hat{\pi}_0 = \underline{\hspace{2cm}}.$$

Then:

$$\hat{x}_0 = 0.01 - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}.$$

Final answers:

$$\hat{\pi}_0 = \underline{\hspace{2cm}}, \quad \hat{x}_0 = \underline{\hspace{2cm}}.$$

Intuition: The Central Bank responds to inflation, so the real interest rate does not fall as much. Therefore the output gap rises only slightly.

Solution

Again, $\tilde{\kappa} = 1.1$. Since $\delta = 1$,

$$\hat{i}_0 = \hat{\pi}_0.$$

The IS curve gives

$$\hat{x}_0 = -(\hat{\pi}_0 - 0.01) = 0.01 - \hat{\pi}_0.$$

The Phillips curve gives

$$\hat{\pi}_0 = 0.9(0.01) + 1.1(0.01 - \hat{\pi}_0) = 0.02 - 1.1\hat{\pi}_0.$$

Therefore,

$$2.1\hat{\pi}_0 = 0.02 \quad \Rightarrow \quad \hat{\pi}_0 = \mathbf{0.0095}.$$

Then

$$\hat{x}_0 = 0.01 - \mathbf{0.0095} = \mathbf{0.0005}.$$

Thus,

$\hat{\pi}_0 = 0.0095, \quad \hat{x}_0 = 0.0005.$

Explanation

The Central Bank responds to inflation. When $\hat{\pi}_0$ rises, $\hat{i}_0$ rises too. This stops the real interest rate from falling too much, so $\hat{x}_0$ rises only slightly.

3.3. Exercise 9.3

Question

Consider

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t, \quad (1.NK^*)$$

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t, \quad (2.NK^*)$$

$$\hat{i}_t = \delta \hat{\pi}_t + v_t. \quad (3B.NK^*)$$

Assume the economy is at steady state in period -1 , $u_t = v_t = 0$, $E_0[\hat{\pi}_1] = 0.01$, $E_0[\hat{x}_1] = 0$, $\beta = 0.9$, $\sigma = 1$, $\kappa = 0.5$. Calculate the δ that makes $\hat{\pi}_0 = 0.005$.

Given:

$$E_0[\hat{\pi}_1] = 0.01, \quad E_0[\hat{x}_1] = 0, \quad u_0 = v_0 = 0, \quad \beta = 0.9, \quad \sigma = 1, \quad \tilde{\kappa} = 1.1.$$

Target:

$$\hat{\pi}_0 = 0.005.$$

Step 1: Use the policy rule.

$$\hat{i}_0 = \delta \hat{\pi}_0.$$

Step 2: Use the IS curve.

$$\hat{x}_0 = -(\delta \hat{\pi}_0 - 0.01) = 0.01 - \delta \hat{\pi}_0.$$

Step 3: Use the Phillips curve.

$$\hat{\pi}_0 = 0.9(0.01) + 1.1(0.01 - \delta \hat{\pi}_0).$$

Simplify:

$$\hat{\pi}_0 = 0.009 + 0.011 - 1.1\delta \hat{\pi}_0.$$

$$\hat{\pi}_0 = 0.02 - 1.1\delta \hat{\pi}_0.$$

Step 4: Rearrange for δ .

$$1.1\delta \hat{\pi}_0 = 0.02 - \hat{\pi}_0.$$

$$\delta = \frac{0.02 - \hat{\pi}_0}{1.1\hat{\pi}_0}.$$

Substitute $\hat{\pi}_0 = 0.005$:

$$\delta = \frac{0.02 - 0.005}{1.1(0.005)} = \underline{\hspace{2cm}}.$$

Step 5: Find $\hat{x}_0$.

$$\hat{x}_0 = 0.01 - \delta \hat{\pi}_0.$$

$$\hat{x}_0 = 0.01 - (\underline{\hspace{2cm}})(0.005) = \underline{\hspace{2cm}}.$$

Final answers:

$$\delta = \underline{\hspace{2cm}}, \quad \hat{x}_0 = \underline{\hspace{2cm}}.$$

Intuition: To make inflation lower, the Central Bank must respond strongly to inflation. A strong response raises the real interest rate and creates a negative output gap.

Solution

Again, $\tilde{\kappa} = 1.1$. The policy rule gives

$$\hat{i}_0 = \delta \hat{\pi}_0.$$

The IS curve gives

$$\hat{x}_0 = -(\delta \hat{\pi}_0 - 0.01) = 0.01 - \delta \hat{\pi}_0.$$

The Phillips curve gives

$$\hat{\pi}_0 = 0.9(0.01) + 1.1(0.01 - \delta \hat{\pi}_0) = 0.02 - 1.1\delta \hat{\pi}_0.$$

Hence,

$$\delta = \frac{0.02 - \hat{\pi}_0}{1.1\hat{\pi}_0}.$$

Set $\hat{\pi}_0 = 0.005$:

$$\delta = \frac{0.02 - 0.005}{1.1(0.005)} = 2.727.$$

Then

$$\hat{x}_0 = 0.01 - 2.727(0.005) = -0.00364.$$

Therefore,

$$\delta = 2.727, \quad \hat{x}_0 = -0.00364.$$

Explanation

To lower inflation, the Central Bank must respond strongly to inflation. A larger δ raises $\hat{i}_0$, raises the real interest rate, lowers demand, and makes $\hat{x}_0 < 0$. This negative output gap lowers inflation.

3.4. Exercise 9.4

Question

Consider

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t, \quad (1.NK^*)$$

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t, \quad (2.NK^*)$$

$$\hat{i}_t = \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t. \quad (3C.NK^*)$$

Assume the economy is at steady state in period -1 , $u_t = v_t = 0$, $E_0[\hat{\pi}_1] = 0.01$, $E_0[\hat{x}_1] = 0$, $\beta = 0.9$, $\sigma = 1$, $\kappa = 0.5$, $\delta_\pi = 1$, and $\delta_x = 1$. Calculate the effect on $\hat{x}_0$ and $\hat{\pi}_0$.

Given:

$$E_0[\hat{\pi}_1] = 0.01, \quad E_0[\hat{x}_1] = 0, \quad u_0 = v_0 = 0, \quad \beta = 0.9, \quad \sigma = 1, \quad \tilde{\kappa} = 1.1, \\ \delta_\pi = 1, \quad \delta_x = 1.$$

Step 1: Use the policy rule.

$$\hat{i}_0 = \delta_\pi \hat{\pi}_0 + \delta_x \hat{x}_0.$$

Since $\delta_\pi = 1$ and $\delta_x = 1$,

$$\hat{i}_0 = \hat{\pi}_0 + \hat{x}_0.$$

Step 2: Use the IS curve.

$$\hat{x}_0 = - \left(\hat{i}_0 - 0.01 \right).$$

Substitute:

$$\hat{x}_0 = - \left(\hat{\pi}_0 + \hat{x}_0 - 0.01 \right).$$

Rearrange:

$$2\hat{x}_0 = 0.01 - \hat{\pi}_0.$$

$$\hat{x}_0 = \frac{1}{2}(0.01 - \hat{\pi}_0).$$

Step 3: Use the Phillips curve.

$$\hat{\pi}_0 = 0.9(0.01) + 1.1\hat{x}_0.$$

Substitute:

$$\hat{\pi}_0 = 0.9(0.01) + 1.1 \left[\frac{1}{2}(0.01 - \hat{\pi}_0) \right].$$

Simplify:

$$\hat{\pi}_0 = 0.009 + 0.55(0.01 - \hat{\pi}_0).$$

$$\hat{\pi}_0 = 0.0145 - 0.55\hat{\pi}_0.$$

Move terms:

$$1.55\hat{\pi}_0 = 0.0145.$$

Solve:

$$\hat{\pi}_0 = \underline{\hspace{2cm}}.$$

Then:

$$\hat{x}_0 = \frac{1}{2}(0.01 - \underline{\hspace{2cm}}) = \underline{\hspace{2cm}}.$$

Final answers:

$$\hat{\pi}_0 = \underline{\hspace{2cm}}, \quad \hat{x}_0 = \underline{\hspace{2cm}}.$$

Intuition: The Central Bank responds to both inflation and the output gap. This makes the interest rate response stronger and reduces the increase in the output gap.

Solution

Again, $\tilde{\kappa} = 1.1$. The policy rule is

$$\hat{i}_0 = \hat{\pi}_0 + \hat{x}_0.$$

The IS curve gives

$$\hat{x}_0 = -(\hat{\pi}_0 + \hat{x}_0 - 0.01) \quad \Rightarrow \quad 2\hat{x}_0 = 0.01 - \hat{\pi}_0.$$

Thus,

$$\hat{x}_0 = \frac{1}{2}(0.01 - \hat{\pi}_0).$$

The Phillips curve gives

$$\hat{\pi}_0 = 0.9(0.01) + 1.1 \left[\frac{1}{2}(0.01 - \hat{\pi}_0) \right] = 0.0145 - 0.55\hat{\pi}_0.$$

Therefore,

$$1.55\hat{\pi}_0 = 0.0145 \quad \Rightarrow \quad \hat{\pi}_0 = \mathbf{0.0093}.$$

Then

$$\hat{x}_0 = \frac{1}{2}(0.01 - \mathbf{0.0093}) = \mathbf{0.0003}.$$

Thus,

$$\boxed{\hat{\pi}_0 = \mathbf{0.0093}, \quad \hat{x}_0 = \mathbf{0.0003}.$$

Explanation

The Central Bank responds to both inflation and the output gap. When expected inflation increases demand, $\hat{x}_0$ rises. But the term $\delta_x \hat{x}_t$ makes the Central Bank raise $\hat{i}_0$, reducing the rise in $\hat{x}_0$ and $\hat{\pi}_0$.

3.5. Exercise 9.5

Question

Consider

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t, \quad (1.NK^*)$$

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t, \quad (2.NK^*)$$

$$\hat{i}_t = \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t. \quad (3C.NK^*)$$

Assume the economy is at steady state in period -1 , $u_t = v_t = 0$, $E_0[\hat{\pi}_1] = 0.01$, $E_0[\hat{x}_1] = 0$, $\beta = 0.9$, $\sigma = 1$, $\kappa = 0.5$, and $\delta_x = 1$. Calculate the δ_π that makes $\hat{\pi}_0 = 0.005$.

Given:

$$E_0[\hat{\pi}_1] = 0.01, \quad E_0[\hat{x}_1] = 0, \quad u_0 = v_0 = 0, \quad \beta = 0.9, \quad \sigma = 1, \quad \tilde{\kappa} = 1.1, \\ \delta_x = 1.$$

Target:

$$\hat{\pi}_0 = 0.005.$$

Step 1: Use the policy rule.

$$\hat{i}_0 = \delta_\pi \hat{\pi}_0 + \delta_x \hat{x}_0.$$

Since $\delta_x = 1$,

$$\hat{i}_0 = \delta_\pi \hat{\pi}_0 + \hat{x}_0.$$

Step 2: Use the IS curve.

$$\hat{x}_0 = -(\delta_\pi \hat{\pi}_0 + \hat{x}_0 - 0.01).$$

Rearrange:

$$2\hat{x}_0 = 0.01 - \delta_\pi \hat{\pi}_0.$$

$$\hat{x}_0 = \frac{1}{2}(0.01 - \delta_\pi \hat{\pi}_0).$$

Step 3: Use the Phillips curve.

$$\hat{\pi}_0 = 0.9(0.01) + 1.1\hat{x}_0.$$

Substitute:

$$\hat{\pi}_0 = 0.009 + 1.1 \left[\frac{1}{2}(0.01 - \delta_\pi \hat{\pi}_0) \right].$$

Simplify:

$$\hat{\pi}_0 = 0.0145 - 0.55\delta_\pi \hat{\pi}_0.$$

Step 4: Use the target $\hat{\pi}_0 = 0.005$.

$$0.005 = 0.0145 - 0.55\delta_\pi(0.005).$$

Rearrange:

$$0.55\delta_\pi(0.005) = 0.0145 - 0.005.$$

$$\delta_\pi = \frac{0.0145 - 0.005}{0.55(0.005)} = \underline{\hspace{2cm}}.$$

Step 5: Find $\hat{x}_0$.

$$\hat{x}_0 = \frac{1}{2} [0.01 - \delta_\pi(0.005)] .$$

$$\hat{x}_0 = \frac{1}{2} [0.01 - (\text{_____})(0.005)] = \text{_____} .$$

Final answers:

$$\delta_\pi = \text{_____}, \quad \hat{x}_0 = \text{_____} .$$

Intuition: To reduce inflation to 0.005, the Central Bank needs a strong response to inflation. This raises the real interest rate and creates a negative output gap.

Solution

The policy rule is

$$\hat{i}_0 = \delta_\pi \hat{\pi}_0 + \hat{x}_0 .$$

The IS curve gives

$$\hat{x}_0 = -(\delta_\pi \hat{\pi}_0 + \hat{x}_0 - 0.01) \quad \Rightarrow \quad 2\hat{x}_0 = 0.01 - \delta_\pi \hat{\pi}_0 .$$

Thus,

$$\hat{x}_0 = \frac{1}{2}(0.01 - \delta_\pi \hat{\pi}_0) .$$

The Phillips curve gives

$$\hat{\pi}_0 = 0.9(0.01) + 1.1 \left[\frac{1}{2}(0.01 - \delta_\pi \hat{\pi}_0) \right] = 0.0145 - 0.55\delta_\pi \hat{\pi}_0 .$$

Set $\hat{\pi}_0 = 0.005$:

$$0.005 = 0.0145 - 0.55\delta_\pi(0.005) .$$

Therefore,

$$\delta_\pi = \frac{0.0145 - 0.005}{0.55(0.005)} = 3.4545 .$$

Then

$$\hat{x}_0 = \frac{1}{2} [0.01 - 3.4545(0.005)] = -0.0036 .$$

Thus,

$$\boxed{\delta_\pi = 3.4545, \quad \hat{x}_0 = -0.0036 .}$$

Explanation

To reduce inflation to 0.005, the Central Bank must respond strongly to inflation. A larger δ_π raises $\hat{i}_0$, raises the real interest rate, reduces demand, and creates a negative output gap.

3.6. Exercise 9.6

Question

Consider

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t, \quad (1.NK^*)$$

$$\hat{\pi}_t = \tilde{\kappa} \sum_{k=0}^{\infty} \beta^k E_t[\hat{x}_{t+k}] + \varepsilon_t^w, \quad (2TO.NK^*)$$

with

$$\tilde{\kappa} = (\sigma + \phi) \frac{(1 - \kappa)(1 - \beta\kappa)}{\kappa}.$$

Let $\beta = 0.99$, $\sigma = 1$, $\kappa = 0.20$, $E_1[\hat{x}_{t+k}] = 0$ for all $k \geq 1$, and $E_1[\hat{\pi}_2] = 0$. Let the economy be at steady state in period 0. Consider a positive cost-push shock $\varepsilon_1^w = 1$ in period 1. There are no other realized or expected shocks. What should the Central Bank do to keep inflation at zero, $\hat{\pi}_1 = 0$? What should it do if instead it wants to stabilize output at $\hat{x}_1 = 0$?

Given:

$$\begin{aligned} \beta &= 0.99, & \sigma &= 1, & \kappa &= 0.20, & \tilde{\kappa} &= 6.416, \\ E_1[\hat{x}_{t+k}] &= 0 \text{ for all } k \geq 1, & E_1[\hat{\pi}_2] &= 0, & \varepsilon_1^w &= 1. \end{aligned}$$

Step 1: Use the IS curve.

$$\hat{x}_1 = E_1[\hat{x}_2] - \frac{1}{\sigma} \left(\hat{i}_1 - E_1[\hat{\pi}_2] \right) + u_1.$$

Substitute:

$$\hat{x}_1 = 0 - \frac{1}{1}(\hat{i}_1 - 0) + 0.$$

So:

$$\hat{x}_1 = -\hat{i}_1.$$

Step 2: Use the trade-off Phillips curve.

$$\hat{\pi}_1 = \tilde{\kappa} \sum_{k=0}^{\infty} \beta^k E_1[\hat{x}_{1+k}] + \varepsilon_1^w.$$

Because future expected output gaps are zero:

$$\hat{\pi}_1 = 6.416\hat{x}_1 + 1.$$

Substitute $\hat{x}_1 = -\hat{i}_1$:

$$\hat{\pi}_1 = -6.416\hat{i}_1 + 1.$$

Case 1: keep inflation at zero.

Set:

$$\hat{\pi}_1 = 0.$$

$$0 = -6.416\hat{i}_1 + 1.$$

Solve:

$$\hat{i}_1 = \frac{1}{6.416} = \underline{\hspace{2cm}}.$$

Then:

$$\hat{x}_1 = -\hat{i}_1 = \underline{\hspace{2cm}}.$$

Case 2: stabilize output at zero.

Set:

$$\hat{x}_1 = 0.$$

Since $\hat{x}_1 = -\hat{i}_1$,

$$\hat{i}_1 = \underline{\hspace{2cm}}.$$

Then:

$$\hat{\pi}_1 = 6.416(0) + 1 = \underline{\hspace{2cm}}.$$

Final answers:

$$\text{If } \hat{\pi}_1 = 0 : \quad \hat{i}_1 = \underline{\hspace{2cm}}, \quad \hat{x}_1 = \underline{\hspace{2cm}}.$$

$$\text{If } \hat{x}_1 = 0 : \quad \hat{i}_1 = \underline{\hspace{2cm}}, \quad \hat{\pi}_1 = \underline{\hspace{2cm}}.$$

Intuition: A cost-push shock raises inflation directly. To keep inflation at zero, the Central Bank must create a negative output gap. If it stabilizes output instead, inflation rises.

Solution

For these parameter values,

$$\tilde{\kappa} = 6.416.$$

Using the IS curve:

$$\hat{x}_1 = E_1[\hat{x}_2] - \frac{1}{\sigma}(\hat{i}_1 - E_1[\hat{\pi}_2]) + u_1.$$

Since $E_1[\hat{x}_2] = 0$, $E_1[\hat{\pi}_2] = 0$, $u_1 = 0$, and $\sigma = 1$,

$$\hat{x}_1 = -\hat{i}_1.$$

The trade-off Phillips curve becomes

$$\hat{\pi}_1 = 6.416\hat{x}_1 + 1.$$

Substitute $\hat{x}_1 = -\hat{i}_1$:

$$\hat{\pi}_1 = -6.416\hat{i}_1 + 1.$$

Case 1: keep inflation at zero. Set $\hat{\pi}_1 = 0$:

$$0 = -6.416\hat{i}_1 + 1 \quad \Rightarrow \quad \hat{i}_1 = \frac{1}{6.416} = 0.15586.$$

Therefore, the Central Bank should increase the interest rate by

15.586%.

Then

$$\hat{x}_1 = -\hat{i}_1 = -0.15586.$$

Case 2: stabilize output at zero. Set $\hat{x}_1 = 0$. Since $\hat{x}_1 = -\hat{i}_1$,

$$\hat{i}_1 = 0.$$

Then

$$\hat{\pi}_1 = 6.416(0) + 1 = \mathbf{1}.$$

Therefore,

$$\hat{i}_1 = 0, \quad \hat{\pi}_1 = 1.$$

Explanation

The positive cost-push shock directly raises inflation. To keep inflation at zero, the Central Bank must create a negative output gap by raising $\hat{i}_1$. But if it stabilizes output at $\hat{x}_1 = 0$, there is no negative output gap to offset the cost-push shock, so inflation rises to 1. This is the monetary policy trade-off:

$$\text{stabilize inflation} \Rightarrow \hat{x}_1 < 0, \quad \text{stabilize output} \Rightarrow \hat{\pi}_1 > 0.$$

4. Main Takeaway

Lecture 7 shows the monetary policy shock:

$$\varepsilon_t \uparrow \Rightarrow v_t \uparrow \Rightarrow \hat{i}_t \uparrow \Rightarrow \hat{x}_t \downarrow \Rightarrow \hat{\pi}_t \downarrow.$$

Lecture 8 studies technology shocks, spending shocks and cost-push shocks:

$$\omega_t, \quad \omega_t^g, \quad \varepsilon_t^w.$$

Lecture 9 shows why monetary policy rules matter. If the Central Bank does not respond to inflation expectations, expectations can become self-fulfilling. If the Central Bank responds strongly enough, inflation can be stabilized. However, with a cost-push shock, stabilizing inflation requires a negative output gap:

monetary policy faces a trade-off between inflation stabilization and output stabilization.